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I declare that I have not cheated, received, or given any assistance by any means throughout this exam. I am aware that violation of these principles according to the University regulations will result in a zero grade and a disciplinary investigation.

1)

a) $f(z) = \ln(z)$

$$f(z) = f(z_0) + \frac{f'(z_0)(z-z_0)}{1} + \frac{f''(z_0)(z-z_0)^2}{2!} + \frac{f^{n}(z_0)}{n!}(z-z_0)^n$$

$z_0 = 1+i$

$$f(z) = \sum_{n=1}^{\infty} (-1)^{(n+1)} \frac{(z-1-i)^n}{n!\,(1+i)^n}$$

b) $z - z_0 = R$ $\qquad R = \sqrt{2}$

$$f(w) = \frac{w^2 - 2w + 2}{w - z}$$

Q4)

$$\text{Res}[f(w), w=z] = \lim_{w \to z} [(w-z) \times f(w)]$$

$$= \lim_{w \to z} \left[(w-z) \times \left(\frac{(w^2 - 2w + 2)}{w - z}\right)\right]$$

$$= \lim_{w \to z} \left[(w^2 - 2w + 2)\right]$$

$$= z^2 - 2z + 2$$

$$f(w) \longrightarrow f(z) = \underline{(z^2 - 2z + 2)}$$

$$\int_C \frac{w^2 - 2w + 2}{w - z} dw = 2\pi i \times \text{Res}[f(w), w=z]$$

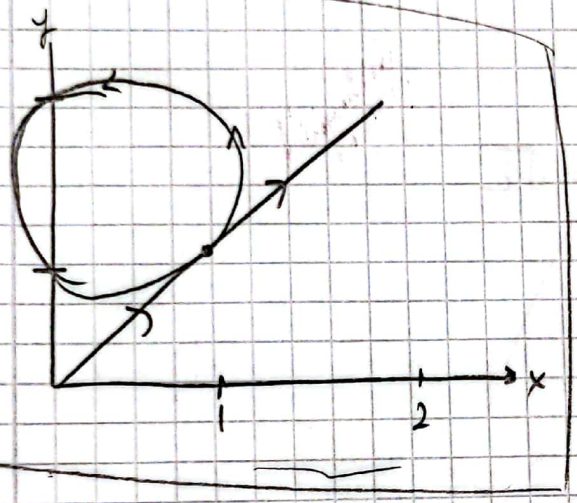
$$= \underline{2\pi i \times (z^2 - 2z + 2)}$$

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9.5) $\int_C e^z dz$ from $z=0$ to $z=2+2i$ — $y=x$ and a circle tangent to line at $(1,1)$

$z = x+iy$, $dz = dx + i\,dy$, $y = x \Rightarrow dy = dx$

$$\rightarrow \int_C e^{(x+iy)}(dx+i\,dy) = \int_C e^{x(1+i)}(1+i)\,dx$$



C is $0 \le x \le 2$

$$= (1+i)\int_0^2 e^{x(1+i)}\,dx$$

$$= \frac{1+i}{1+i}\left(e^{x(1+i)}\right)\Big|_0^2$$

$$= \left[e^{2(1+i)} - e^{0(1+i)}\right]$$

$$= e^{2(1+i)} - 1$$

6) $f(z) = \frac{3z-1}{z(z-1)}$

$f(z) = 0 \rightarrow z = 0$ and $z = 1$

$$\int \frac{3z-1}{z(z-1)}\,dz = \underbrace{\int \frac{\frac{3z-1}{(z-1)}}{z}\,dz}_{z=0 \text{ (clockwise)}} + \underbrace{\int \frac{\frac{3z-1}{z}}{z-1}\,dz}_{z=1 \text{ (counterclockwise)}}$$

$= 1$

$f(z) = \frac{3z-1}{z-1}$ $\qquad$ $f(z) = \frac{3z-1}{z}$

$$\int \frac{\frac{3z-1}{z-1}}{z}\,dz \quad (z=0) = f(z)\cdot(2\pi i) = \underbrace{f(0)}_{1}\cdot 2\pi i \rightarrow 2\pi i^{\circ} \text{ clockwise}$$

$$\int \frac{\frac{3z-1}{z}}{z-1}\,dz \quad (z=1) = f(z)\cdot(2\pi i) = \underbrace{f(1)}_{2}\cdot(2\pi i) \rightarrow 4\pi i^{\circ} \text{ counterclockwise}$$

$4\pi i^{\circ} - 2\pi i^{\circ} = \underline{2\pi i^{\circ}}$

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